

Validity Analysis: DRBENCHMARK

Target context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Overall risk: **HIGH**

Deployment Context

Use case: In a hospital setting, a clinician uses software to manage unstructured French clinical cases by applying a model evaluated on part-of-speech tagging, named-entity recognition, and multi-label classification. The system identifies specific clinical and temporal entities while mapping pathologies to standardized medical classification chapters and axes to automatically generate a patient's clinical profile. These outputs directly help the clinician summarize complex medical histories and prioritize care based on the severity of identified symptoms, ensuring that life-critical decisions are supported by accurately synthesized data.

Target population: French-speaking medical professionals, including clinicians, hospital physicians, and nursing staff in French-speaking health systems.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content △	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology △	2	Significant gaps	high
Output Content △	2	Significant gaps	high
Output Form △	2	Significant gaps	high
Average	2.5		

△ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

DrBenchmark is the most comprehensive French biomedical NLP benchmark available and provides genuine partial coverage for the Metropolitan French hospital clinical NLP deployment, particularly through DEFT-2021 (multi-label MeSH classification + rich NER with temporal entities), CAS and CLISTER (clinical-case French), and DiaMed (only ICD-10 chapter-level labels

with documented IAA and medical-expert involvement). Input Form is well aligned (text-only standard French, Latin diacritics, HuggingFace toolkit). However, four of the six dimensions show significant validity concerns for this deployment: Input Content (score 2) — no raw-EHR register, substantial non-metropolitan content in DiaMed and MorFITT, register systematically cleaner than active hospital notes; Output Ontology (score 2) — no confidence-scored / ranked-candidate / uncertainty-flagging output structure, ICD-10 chapter granularity coarser than operational PMSI/GHM codes, several datasets with orthogonal labels (negation/speculation, pharmacy MCQA); Output Content (score 2) — no documented PMSI coder or hospital coding specialist involvement in any constituent dataset, automatic silver-standard POS annotations in CAS/ESSAI, annotation completeness concerns in E3C and ESSAI; Output Form (score 2) — standard F1/Spearman/SeqEval cannot measure the deployment's required confidence calibration and uncertainty flagging behavior, severe class imbalance in CAS/ESSAI/PxCorpus masks minority-class performance. Input Ontology scores 3 due to broad task coverage but with several irrelevant categories (pharmacy exam, veterinary, encyclopedic) consuming evaluation weight. The deployment's elicitation-flagged HIGH-priority concerns (IC, OO, OC) all map to validity scores of 2.

Practical Guidance

What This Benchmark Measures

DrBenchmark measures core French biomedical language understanding capabilities — POS tagging, clinical and temporal NER, multi-label clinical-profile classification, semantic textual similarity, and intent classification — across 8 state-of-the-art masked language models with reproducible 4-run averaged metrics. For the deployment, the strongest constructs measured are: (1) French clinical NER for pathology, anatomy, treatment, dosage, and temporal entities (via DEFT-2021's 14-type scheme and E3C temporal annotations); (2) multi-label classification capacity across MeSH and ICD-10 chapter axes (via DEFT-2021 and DiaMed); (3) POS tagging on French clinical case text (via CAS).

Construct Depth

Construct depth is shallow-to-moderate for the deployment's core functions. Input Form (4) and Input Ontology (3) provide reasonable construct coverage, but the four dimensions most consequential to the deployment — Input Content, Output Ontology, Output Content, Output Form — all score 2. The benchmark provides binary or single-label evaluation only; the deployment's required confidence-scored multi-label outputs with uncertainty flagging is structurally unmeasurable. Annotation authority for the deployment-critical ICD-10/PMSI mapping function rests on a single medical expert in one non-metropolitan dataset (DiaMed). Aggregate scores conflate clinically relevant tasks with off-domain content (pharmacy exam, veterinary, encyclopedic STS pairs).

What Else You Need

Three concrete supplementation streams are needed before deployment-relevant evaluation conclusions can be drawn: (1) Custom calibration evaluation — calibration curves, ECE, Brier scores, per-label AUROC, and ranked-candidate quality metrics on DEFT-2021 and DiaMed multi-label outputs (addresses OO/OF gaps); (2) Re-annotation by certified PMSI coders on a held-out metropolitan French hospital subset to validate label authority for ICD-10/PMSI mapping (addresses OC gap and the elicitation-flagged annotator-vs-user gap); (3) Construction or curation of a moderate-noise active-hospital-note evaluation set with French clinical abbreviations (ATCD, TTT, HDJ) and dictation artifacts to assess register-transfer performance (addresses IC gap, since web search confirms no French raw-EHR benchmark exists [WEB-1, WEB-18]). Aggregate DrBenchmark scores should be decomposed task-by-task and filtered to exclude FrenchMedMCQA, MorFITT veterinary content, and DEFT-2020 off-domain encyclopedic pairs when interpreting deployment-relevant signal.

ID	Evidence
WEB-1	EHRNoteQA (NeurIPS 2024) demonstrates that existing clinical NLP benchmarks typically fail to represent raw multi-note EHR documentation; no equivalent French benchmark identified (https://arxiv.org/abs/2402.16040)
WEB-18	Recent 2025 meta-analysis confirms a significant proportion of medical LLM benchmarks are constructed from academic materials rather than authentic clinical data — applicable to DrBenchmark's source composition (https://arxiv.org/html/2508.04325)

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

DrBenchmark's task taxonomy spans 20 tasks across POS, NER, MCQA, MCC, MLC, intent classification, and STS [Q1, Q4, Q18, Q81], covering most of the deployment's required functions: POS tagging (CAS, ESSAI), NER with clinical/temporal/anatomical/treatment entities (DEFT-2021, E3C, QUAERO), and multi-label pathology classification (DEFT-2021 MeSH, DiaMed ICD-10 chapters, MorFITT specialties). However, several ontological gaps directly affect the deployment. First, the elicitation flagged tropical/DOM-TOM pathology as a HIGH-priority future extension; while the deployment scope is Metropolitan France only, DiaMed's source (Pan African Medical Journal) introduces sub-Saharan/North African case content that does not match the metropolitan target population (DIAMED-D7, DIAMED-D8, DIAMED-D9). Second, MorFITT's 12 specialty labels and CAS/ESSAI's negation/speculation labels do not map to ICD-10/PMSI diagnostic axes (MORFITT-D11, CAS-D19, ESSAI-D9), introducing construct-irrelevant variance into aggregate benchmark scores. Third, FrenchMedMCQA's pharmacy-exam content (analytical chemistry, electroanalysis, health economics) is largely irrelevant to hospital clinical NLP (FRENCHMEDMCQA-D2, FRENCHMEDMCQA-D23, FRENCHMEDMCQA-D43, FRENCHMEDMCQA-D45). Fourth, the deployment requires confidence-scored multi-label outputs and uncertainty flagging, but no benchmark task category targets confidence-calibrated or ranked-candidate outputs [Q56]. The taxonomy partially aligns with metropolitan deployment needs but contains both omissions (PMSI-granular codes, calibration tasks) and irrelevant categories (veterinary, pharmacy exam, encyclopedic content).

ID	Evidence
Q1	"It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification." (p.1)
Q4	"These tasks encompass part-of-speech (POS) tagging, named-entity recognition (NER), classification, question-answering (QA), and semantic textual similarity (STS)." (p.1)
Q18	"A variety of tasks with different requirements and objectives: Part-of-Speech (POS) tagging, Multi-class, Multi-label and Intent classification, Named-Entity Recognition (NER), Multiple-Choice Question-Answering (MCQA), and Semantic Textual Similarity (STS)." (p.2)
Q56	"The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions." (p.6)
Q81	"Table 8 summarizes the results obtained on average by the considered MLMs when aggregating the tasks into one of the five designated categories: POS, NER, MCQA, MCC (Multi-class classification), MLC (Multi-label classification), or STS tasks." (p.8)
FRENCHMED MCQA-D2	La loi de Beer-Lambert: S'applique aux dosages spectrophotométriques... Est une loi additive — Analytical chemistry question irrelevant to clinical NLP
DIAMED-D7	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabe hospital origin — non-metropolitan French clinical setting
DIAMED-D8	À propos d'un cas clinique ... keywords: ['Niamey'] — Niger-based case — sub-Saharan African clinical context, not metropolitan France
DIAMED-D9	keywords: ['Rate', 'abdomen aigu', 'scanner abdominal', 'chirurgie', 'Maroc'] — Moroccan clinical case — North African origin rather than French metropolitan
ESSAI-D9	Essai clinique d'immunothérapie évaluant l'association du galunisertib (un facteur de croissance transformant β), avec l'anticorps anti-PD-L1 durvalumab (MEDI4736), dans le cancer du pancréas métastatique — Rich pathology content labeled only for negation/speculation, not ICD-10 — output ontology mismatch
MORFITT-D11	Nous rapportons le cas d'une patiente de 71 ans ayant développé un pseudomyxome péritonéal secondaire à un adénocarcinome appendiculaire. — Clinical case abstract with surgery+etiology labels; labels do not map to ICD-10 codes

CAS-D19	label_names: ["negation_speculation", "negation", "neutral", "speculation"] — Classification labels measure linguistic modality, not ICD-10 pathology categories
FRENCHMED MCQA-D23	En électrophorèse capillaire haute performance, le sens de migration de l'analyse dépend: De la nature de la charge de l'analyte — Analytical chemistry instrumentation; not relevant to clinical NLP
FRENCHMED MCQA-D43	L'économie médicale est une économie... Caractérisée par le libre choix des médecins par le malade — Health economics question; low clinical NLP relevance
FRENCHMED MCQA-D45	Parmi les techniques voltampérométriques, on trouve: La polarographie — Electroanalytical chemistry; irrelevant to hospital clinical NLP

Strengths

- Native multi-label classification structure in DEFT-2021 with up to 7 concurrent MeSH labels per case directly exercises multi-morbid pathology classification (DEFT2021-D1, DEFT2021-D2)
- DEFT-2021 NER scheme covers PATHOLOGY, TREATMENT, ANATOMY, DATE, DURATION, FREQUENCY, MOMENT — full alignment with deployment's clinical and temporal entity extraction needs (DEFT2021-D3, DEFT2021-D4, DEFT2021-D5)
- Breadth of 20 tasks ensures most deployment functions (POS, NER, MLC, temporal extraction) have at least one structurally aligned dataset

ID	Evidence
DEFT2021-D1	Un homme de 42 ans était hospitalisé en urgence le 10 décembre 2003 dans le service de gastro-entérologie pour un tableau d'hépatite aiguë — Multi-morbid hepatology/toxicology/psychiatry case with 6 concurrent MeSH labels
DEFT2021-D2	Nous rapportons le cas d'un garçon de huit ans [...] microangiopathie thrombotique secondaire à un syndrome hémolytique urémique (SHU) atypique — 7-label pediatric multi-morbid case illustrating complex label overlap
DEFT2021-D3	cotrimoxazole pour 10 jours — Treatment drug with duration span annotated — confirms temporal treatment labeling
DEFT2021-D4	Hydrocortisone 300 mg IV immédiatement ; Traitement Hydroxyzine 25 mg par voie orale toutes les 6 heures — Multiple entity types co-occurring in a single sentence, confirming NER scheme coverage
DEFT2021-D5	Au jour 42 , une deuxième désensibilisation a eu lieu — Temporal anchor (relative day) labeled, confirming temporal entity extraction alignment

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required deployment categories include: POS tagging on clinical text, clinical-entity NER (pathology, anatomy, treatment, dosage), temporal-entity NER (date, duration, frequency, moment), multi-label pathology classification mapped to ICD-10/PMSI, and confidence-scored ranked-candidate output. DrBenchmark covers POS (CAS, ESSAI) [Q42, Q43], clinical NER [Q108], temporal NER [Q104, Q108, DEFT2021-D3, DEFT2021-D5], and multi-label classification [Q26, Q27, Q45, DEFT2021-D1].

IO-2: Confidence-scored / ranked-candidate / uncertainty-flagged output categories are absent — the benchmark uses standard classification metrics and majority-class baselines [Q56]. PMSI-granular code classification is also absent; DiaMed only reaches ICD-10 chapter level (DIAMED-D11). Tropical/DOM-TOM pathology categories not present in French clinical subsets (E3C-D4, E3C-D13).

IO-3: FrenchMedMCQA pharmacy exam content covering electroanalysis, analytical chemistry, and health economics is largely irrelevant to clinical NLP deployment (FRENCHMEDMCQA-D23, FRENCHMEDMCQA-D43, FRENCHMEDMCQA-D45). MorFITT contains substantial veterinary content (equine, porcine, canine) off-domain for human hospital deployment (MORFITT-D5, MORFITT-D9, MORFITT-D26). DEFT-2020 contains 30–40% off-domain encyclopedic content (railways, video games, sports) (DEFT2020-D9, DEFT2020-D10, DEFT2020-D11, DEFT2020-D12). CAS/ESSAI negation/speculation classification labels are linguistically rather than clinically relevant to ICD-10 mapping (CAS-D19, ESSAI-D9).

IO-4: Construct underrepresentation: confidence calibration, PMSI-granular codes, DOM-TOM tropical pathology (out of immediate scope but flagged priority extension). Construct-irrelevant variance: pharmacy exam content, veterinary content, encyclopedic STS pairs. Aggregate scores across 20 tasks may obscure deployment-relevant performance.

ID	Evidence
Q1	"It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification." (p.1)
Q18	"A variety of tasks with different requirements and objectives: Part-of-Speech (POS) tagging, Multi-class, Multi-label and Intent classification, Named-Entity Recognition (NER), Multiple-Choice Question-Answering (MCQA), and Semantic Textual Similarity (STS)." (p.2)
Q26	"The multi-label classification task focuses on identifying the patient's clinical profile based on the diseases, signs, or symptoms mentioned in the clinical cases." (p.3)
Q27	"The dataset is annotated with 23 axes from Chapter C of the Medical Subject Headings (MeSH)." (p.3)
Q39	"FrenchMedMCQA (Labrak et al., 2022) is a Multiple-Choice Question-Answering (MCQA) dataset for biomedical domain. It contains 3,105 questions coming from real exams of the French medical specialization diploma in pharmacy, integrating single and multiple answers. The first task consists of automatically identifying the set of correct answers among the 5 proposed for a given question. The second task consists of identifying the number of answers (between 1 and 5) supposedly correct for a given question." (p.4)
Q42	"CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)
Q43	"ESSAI (Dalloux et al., 2021) contains 7,247 clinical trial protocols annotated in 41 POS tags using TreeTagger (Schmid, 1994). As the dataset was not originally divided into 3 subsets, we applied the same procedure as on the CAS corpus." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
Q56	"The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions." (p.6)
Q104	"Clinical: O, and CLINENTITY. Temporal: O, EVENT, ACTOR, BODYPART, TIMEX3 and RML" (p.14)
Q108	"Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE" (p.14)
DEFT2021-D1	Un homme de 42 ans était hospitalisé en urgence le 10 décembre 2003 dans le service de gastro-entérologie pour un tableau d'hépatite aiguë — Multi-morbid hepatology/toxicology/psychiatry case with 6 concurrent MeSH labels
DEFT2021-D3	cotrimoxazole pour 10 jours — Treatment drug with duration span annotated — confirms temporal treatment labeling
E3C-D4	The presence of terminal spine confirmed Schistosoma haematobium and a diagnosis of schistosomiasis was made. — Tropical parasitic pathology in English config — confirms tropical content structurally absent from French subset
DEFT2021-D5	Au jour 42 , une deuxième désensibilisation a eu lieu — Temporal anchor (relative day) labeled, confirming temporal entity extraction alignment
MORFITT-D5	seules des données limitées sont disponibles sur le surfactant chez le cheval... notre objectif était de mener une étude immunocytochimique au microscope optique et électronique sur les poumons normaux du cheval. — Equine veterinary immunology study; off-domain for human hospital deployment
MORFITT-D9	L'otite externe est une maladie multifactorielle fréquente chez le chien... Décrire le microbiote bactérien auriculaire des chiens avec otite externe comparé aux chiens sains. — Canine ear microbiome study; veterinary content off-domain for hospital NLP

DEFT2020-D9	Entre Perpignan et Villefranche, il subsiste de très nombreux poteaux caténaires datant des premiers essais en 12 KV 16 2/3 Hz... — Railway infrastructure content; entirely off-domain for clinical NLP deployment
ESSAI-D9	Essai clinique d'immunothérapie évaluant l'association du galunisertib (un facteur de croissance transformant β), avec l'anticorps anti-PD-L1 durvalumab (MEDI4736), dans le cancer du pancréas métastatique — Rich pathology content labeled only for negation/speculation, not ICD-10 — output ontology mismatch
DEFT2020-D10	Boris Fiodorovitch Godounov, en russe : Борис Фёдорович Годунов (v.1551–Moscou, 13 avril 1605)... — Russian historical figure biography; off-domain and contains Cyrillic script
DIAMED-D11	Maladie de Kaposi à localisation broncho-pulmonaire révélant une infection VIH — Chapter-level label only; PMSI granular codes (e.g., B21.x) not evaluated
DEFT2020-D11	Les Canadiens de Montréal sont une franchise de hockey sur glace professionnel située à Montréal dans la province de Québec au Canada. — Professional hockey franchise; irrelevant to hospital deployment
DEFT2020-D12	Lancement du célèbre MMORPG : World of Warcraft. — Video game launch event; entirely off-domain
E3C-D13	a diagnosis of schistosomiasis was made — Tropical disease in English config only; absent from French subset — confirms DOM-TOM pathology gap
CAS-D19	label_names: ["negation_speculation", "negation", "neutral", "speculation"] — Classification labels measure linguistic modality, not ICD-10 pathology categories
FRENCHMED MCQA-D23	En électrophorèse capillaire haute performance, le sens de migration de l'analyse dépend: De la nature de la charge de l'analyte — Analytical chemistry instrumentation; not relevant to clinical NLP
MORFITT-D26	Prévalence et caractéristiques morphologiques et moléculaires de Sarcocystis bertrami chez les chevaux en Chine. — Horse parasitology study from China; off-domain for human hospital deployment
FRENCHMED MCQA-D43	L'économie médicale est une économie... Caractérisée par le libre choix des médecins par le malade — Health economics question; low clinical NLP relevance
FRENCHMED MCQA-D45	Parmi les techniques voltampérométriques, on trouve: La polarographie — Electroanalytical chemistry; irrelevant to hospital clinical NLP
WEB-2	PMSI uses GHM groupings derived from CIM-10 FR principal/secondary diagnoses + CCAM procedures + age + length of stay — substantially more granular than ICD-10 chapter labels (https://www.lespmsi.com/2024/manuel_GHM_2024_volume1_provisoire.pdf)

Information Gaps

- Whether aggregate benchmark scores can be decomposed to filter out off-domain content (veterinary, encyclopedic, pharmacy exam) for deployment-relevant interpretation

Requires Expert Verification

- Whether DEFT-2021's 23 MeSH axes provide adequate proxy for PMSI Catégorie Majeure groupings used in operational hospital coding

Remediation

Gap 1: Several DrBenchmark tasks contribute construct-irrelevant variance to aggregate scores — FrenchMedMCQA pharmacy exam (chemistry/economics), MorFITT veterinary content, DEFT-2020 encyclopedic pairs

Recommendation: When interpreting DrBenchmark performance for this deployment, decompose aggregate scores task-by-task and weight DEFT-2021 (NER + MLC), CAS (POS + clinical case register), CLISTER (clinical STS), and E3C French_temporal as deployment-relevant; treat FrenchMedMCQA, MorFITT veterinary subsets, and DEFT-2020 encyclopedic content as low-relevance signal for aggregate interpretation.

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input Content was flagged HIGH priority in elicitation due to the deployment's requirement to handle moderately noisy hospital notes with abbreviations and shorthand. The benchmark provides standard-French biomedical text [\[Q19\]](#) across diverse sources, which superficially aligns with the deployment language. However, multiple critical concerns surface. First, no DrBenchmark dataset uses raw hospital EHR notes; sources are clinical trial protocols (ESSAI) [\[Q43\]](#), competition corpora (DEFT-2021, DEFT-2020) [\[Q24\]](#), peer-reviewed published case reports (CAS, CLISTER, E3C, DiaMed) [\[Q42, Q41, Q29, Q45\]](#), drug regulatory documents (QUAERO-EMEA) [\[Q34\]](#), MEDLINE titles, and patent claims. Web-source confirmation: no French raw-EHR clinical NLP benchmark exists [\[WEB-1, WEB-18\]](#). Examples of formality gap include CAS-D20, E3C-D9, DEFT2021-D14, DIAMED-D12. Second, substantial non-Metropolitan content contaminates several datasets: DiaMed's cases originate from Burkina Faso, Morocco, and Niger (DIAMED-D7, DIAMED-D8, DIAMED-D9); MorFITT contains Canadian, Jordanian, Saudi, and Arabic-population research content (MORFITT-D10, MORFITT-D20, MORFITT-D22, MORFITT-D31); E3C non-French configs (Basque, English, Italian, Spanish) dominate the multilingual structure (E3C-D7, E3C-D8). Third, DEFT-2020 contains 30–40% off-domain encyclopedic material (DEFT2020-D9, DEFT2020-D10, DEFT2020-D11). Fourth, PxCorpus — the closest noise proxy — contains system-prompt contamination corrupting clinical register (PXCORPUS-D22, PXCORPUS-D23). The CAS, CLISTER, DEFT-2021, and ESSAI corpora provide partial metropolitan-French clinical signal with quantified lab values and standard medical terminology (CAS-D2, CLISTER-D2, DEFT2021-D7), but all are edited published materials with cleaner register than active hospital notes (CAS-D20).

ID	Evidence
Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
Q24	"DEFT-2021 (Grouin et al., 2021) is a subset of 275 clinical cases taken from the 2019 edition of DEFT." (p.3)
Q29	"E3C (Magnini et al., 2020) is a multilingual dataset of clinical cases annotated for the NER task." (p.3)
Q34	"The dataset covers two text genres (drug leaflets and biomedical titles), consisting of a total of 103,056 words sourced from EMEA or MEDLINE." (p.3)
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
Q42	"CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)
Q43	"ESSAI (Dalloux et al., 2021) contains 7,247 clinical trial protocols annotated in 41 POS tags using TreeTagger (Schmid, 1994). As the dataset was not originally divided into 3 subsets, we applied the same procedure as on the CAS corpus." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)

CAS-D2	“les constantes hémodynamiques étaient stables mais le bilan biologique mettait en évidence une anémie à 5 , 7 g / 100ml d' hb , une hyponatrémie à 128 meq / l , une acidose métabolique...” — Dense lab-value notation typical of deployment's severity signal extraction
CLISTER-D2	Le bilan rénal a objectivé une insuffisance rénale avec une créatininémie à 1440 µmol/l, soit une clairance de la créatinine à 6,7 ml/mn/m2, urémie à 2,07 g/l. — Quantified biological workup with clinical abbreviations; representative of standard hospital documentation
DIAMED-D7	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabe hospital origin — non-metropolitan French clinical setting
E3C-D7	Azken hilabeteetan lkernek kodeina + ibuprofenoa hartu ditu, baina ez du hobekuntza handirik nabaritu. — Basque language — entirely irrelevant to French deployment context
DEFT2021-D7	bilan biologique d'entrée révélait une macrocytose avec une thrombopénie, un cytolyse majeure avec des ALAT et des ASAT respectivement à 60 à 450 fois la normale — Dense laboratory values with quantified severity signals
DIAMED-D8	À propos d'un cas clinique ... keywords: ['Niamey'] — Niger-based case — sub-Saharan African clinical context, not metropolitan France
E3C-D8	Aurrekariak: - Krisi epileptikoa 12 urte zituenean, bere herrialdean tratatua: BOLIBIA. — Basque text referencing Bolivia — not relevant to French metropolitan hospital NLP
E3C-D9	Le bilan biologique montrait une cholestase (bilirubine totale a ` 140 mmol/L... — Well-formed published case report prose — contrasts with noisy active-hospital EHR shorthand
DIAMED-D9	keywords: ['Rate', 'abdomen aigu', 'scanner abdominal', 'chirurgie', 'Maroc'] — Moroccan clinical case — North African origin rather than French metropolitan
DEFT2020-D9	Entre Perpignan et Villefranche, il subsiste de très nombreux poteaux caténaires datant des premiers essais en 12 KV 16 2/3 Hz... — Railway infrastructure content; entirely off-domain for clinical NLP deployment
MORFITT-D10	L'objectif de cette étude était de documenter les pratiques actuelles de gestion des maladies dans les exploitations apicoles situées au sud-ouest du Québec, Canada. — Canadian beekeeping disease management study; non-Metropolitan French origin
DEFT2020-D10	Boris Fiodorovitch Godounov, en russe : Борис Фёдорович Годунов (v.1551–Moscou, 13 avril 1605)... — Russian historical figure biography; off-domain and contains Cyrillic script
DEFT2020-D11	Les Canadiens de Montréal sont une franchise de hockey sur glace professionnel située à Montréal dans la province de Québec au Canada. — Professional hockey franchise; irrelevant to hospital deployment
DIAMED-D12	L'examen clinique révélait une paralysie flasque aréflexique avec amyotrophie importante des membres inférieurs. On notait la présence de quelques trépidations épileptoïdes du pied gauche. — Polished academic prose — formal register unlike noisy active-hospital EHR notes
DEFT2021-D14	L' ECBU était stérile . — Standard abbreviation in complete sentence — published case report register, not raw EHR shorthand
CAS-D20	“vous êtes appelés au secours d' une infirmière de nuit pour confusion chez un patient bronchopathe chronique , tabagique non sevré à 70 paquets-années .” — Narrative clinical description; more formal than raw EHR shorthand
MORFITT-D20	79 patients au total ont été recrutés à Amman (Jordanie) en 2015. — Jordanian patient population; non-Metropolitan French clinical context
MORFITT-D22	le questionnaire YFAS 2.0-A... peut être utilisée comme équivalent du questionnaire anglais YFAS 2.0 pour étudier l'addiction à la nourriture dans les populations arabophones. — Arabic-population food addiction validation study; non-French population
PXCORPUS-D 22	teralithe 250 milligrammes / le serveur de dialogue met beaucoup de temps à comprendre votre énoncé veuillez reformuler différemment teralithe 250 milligrammes 2 comprimés matin et soir pendant 1 mois — System prompt embedded in prescription utterance; corrupts register
PXCORPUS-D 23	le médicament mais pas le bon le système ne m'a pas compris le bon médicament et le 10 antalvic en gélules — User commentary about ASR failure; non-clinical register
MORFITT-D31	Un autoquestionnaire envoyé par courrier électronique a été rempli par 191 médecins internes sur 207 travaillant dans différentes spécialités et régions de Riyad. — Saudi Arabian intern survey; non-Metropolitan French context
WEB-1	EHRNoteQA (NeurIPS 2024) demonstrates that existing clinical NLP benchmarks typically fail to represent raw multi-note EHR documentation; no equivalent French benchmark identified (https://arxiv.org/abs/2402.16040)
WEB-18	Recent 2025 meta-analysis confirms a significant proportion of medical LLM benchmarks are constructed from academic materials rather than authentic clinical data — applicable to DrBenchmark's source composition (https://arxiv.org/html/2508.04325)

Strengths

- CAS, CLISTER, and DEFT-2021 provide authentic French clinical case prose with quantified lab values, surgical narratives, and diagnostic reasoning closely matching standard hospital documentation register (CAS-D2, CLISTER-D2,

DEFT2021-D7)

- Explicit diversity of data origins — scientific literature, clinical trials, clinical cases, speech transcriptions [Q19] — provides cross-register robustness signal
- PxCorpus offers the benchmark's only naturally noisy register (transcribed speech with hesitations, truncations, ASR artifacts) (PXCORPUS-D1, PXCORPUS-D2, PXCORPUS-D4)

ID	Evidence
Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
PXCORPUS-D1	euh 2 le matin 2 le midi et 2 le soir — Spoken filler in prescription; illustrates non-formal register
CAS-D2	"les constantes hémodynamiques étaient stables mais le bilan biologique mettait en évidence une anémie à 5 , 7 g / 100ml d' hb , une hyponatrémie à 128 meq / l , une acidose métabolique..." — Dense lab-value notation typical of deployment's severity signal extraction
CLISTER-D2	Le bilan rénal a objectivé une insuffisance rénale avec une créatininémie à 1440 µmol/l, soit une clairance de la créatinine à 6,7 ml/mn/m2, urémie à 2,07 g/l. — Quantified biological workup with clinical abbreviations; representative of standard hospital documentation
PXCORPUS-D2	lamotrigine 25 milligrammes euh p/ combien — Incomplete utterance with filler and truncated word; spoken transcription noise
PXCORPUS-D4	ne pas tenir compte à midi tous les jours merd/ — Truncated word at end; raw transcription behavior
DEFT2021-D7	bilan biologique d'entrée révélait une macrocytose avec une thrombopénie, un cytolysse majeure avec des ALAT et des ASAT respectivement à 60 à 450 fois la normale — Dense laboratory values with quantified severity signals

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Inputs are in standard medical French and largely match the deployment's primary language. However, the deployment processes notes with French clinical abbreviations (ATCD, TTT, HDJ) and dictation artifacts that are absent or rare in benchmark sources. Region-specific PMSI/CIM-10 FR coding conventions [WEB-2] are not represented in the source text register.

IC-2: Substantial non-Metropolitan French clinical content is mixed into the benchmark. DiaMed cases are sub-Saharan/North African (DIAMED-D7, DIAMED-D8, DIAMED-D9); MorFITT contains Canadian (Quebec), Jordanian, Saudi, Arabic-population research (MORFITT-D10, MORFITT-D20, MORFITT-D22, MORFITT-D31); DEFT2021 contains some Quebec French terminology (DEFT2021-D20). This may distort performance estimates for the metropolitan target population.

IC-3: The benchmark does not contain Western-specific knowledge unsuitable to the metropolitan French context per se; however, DEFT-2020 contains substantial off-domain encyclopedic content (railways, video games, Russian historical figures, Japanese script) that introduces construct-irrelevant variance (DEFT2020-D9, DEFT2020-D10, DEFT2020-D12, DEFT2020-D13). MorFITT veterinary content is similarly off-domain (MORFITT-D5, MORFITT-D9).

IC-4: INSUFFICIENT DOCUMENTATION — would require recruitment of metropolitan French hospital clinicians and PMSI coders to assess whether benchmark content matches the moderate-noise register of active EHR notes.

IC-5: Critical content gaps: (1) no raw EHR notes [WEB-1, WEB-18]; (2) non-metropolitan content in DiaMed and MorFITT misrepresents the deployment population; (3) edited register systematically cleaner than active hospital documentation (CAS-D20, DEFT2021-D14, DIAMED-D12); (4) French clinical abbreviations (ATCD, TTT, HDJ) under-represented. These collectively harm content validity for the deployment.

ID	Evidence
Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
Q42	"CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
CAS-D2	"les constantes hémodynamiques étaient stables mais le bilan biologique mettait en évidence une anémie à 5 , 7 g / 100ml d' hb , une hyponatrémie à 128 meq / l , une acidose métabolique..." — Dense lab-value notation typical of deployment's severity signal extraction
MORFITT-D5	seules des données limitées sont disponibles sur le surfactant chez le cheval... notre objectif était de mener une étude immunocytochimique au microscope optique et électronique sur les poumons normaux du cheval. — Equine veterinary immunology study; off-domain for human hospital deployment
DIAMED-D7	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabe hospital origin — non-metropolitan French clinical setting
DIAMED-D8	À propos d'un cas clinique ... keywords: ['Niamey'] — Niger-based case — sub-Saharan African clinical context, not metropolitan France
DIAMED-D9	keywords: ['Rate', 'abdomen aigu', 'scanner abdominal', 'chirurgie', 'Maroc'] — Moroccan clinical case — North African origin rather than French metropolitan
DEFT2020-D9	Entre Perpignan et Villefranche, il subsiste de très nombreux poteaux caténaux datant des premiers essais en 12 KV 16 2/3 Hz... — Railway infrastructure content; entirely off-domain for clinical NLP deployment
MORFITT-D9	L'otite externe est une maladie multifactorielle fréquente chez le chien... Décrire le microbiote bactérien auriculaire des chiens avec otite externe comparé aux chiens sains. — Canine ear microbiome study; veterinary content off-domain for hospital NLP
MORFITT-D10	L'objectif de cette étude était de documenter les pratiques actuelles de gestion des maladies dans les exploitations apicoles situées au sud-ouest du Québec, Canada. — Canadian beekeeping disease management study; non-Metropolitan French origin
DEFT2020-D10	Boris Fiodorovitch Godounov, en russe : Борис Фёдорович Годунов (v.1551–Moscou, 13 avril 1605)... — Russian historical figure biography; off-domain and contains Cyrillic script
DEFT2020-D12	Lancement du célèbre MMORPG : World of Warcraft. — Video game launch event; entirely off-domain
DIAMED-D12	L'examen clinique révélait une paralysie flasque aréflexique avec amyotrophie importante des membres inférieurs. On notait la présence de quelques trépidations épileptoïdes du pied gauche. — Polished academic prose — formal register unlike noisy active-hospital EHR notes
DEFT2020-D13	42 Jeux indémodables (だれでもアソビ大全 , Daredemo Asobi Taizen au Japon...) — Japanese script embedded in French text; video game content; off-domain
DEFT2021-D14	L' ECU était stérile . — Standard abbreviation in complete sentence — published case report register, not raw EHR shorthand
MORFITT-D20	79 patients au total ont été recrutés à Amman (Jordanie) en 2015. — Jordanian patient population; non-Metropolitan French clinical context
DEFT2021-D20	Au jour 0, le jeune patient reçoit de l'acétaminophène 300 mg par voie orale — "Acétaminophène" (Quebec French) vs. "paracétamol" (metropolitan French) — minor regional lexical divergence
CAS-D20	"vous êtes appelés au secours d' une infirmière de nuit pour confusion chez un patient bronchopathe chronique , tabagique non sevré à 70 paquets-années ." — Narrative clinical description; more formal than raw EHR shorthand
MORFITT-D22	le questionnaire YFAS 2.0-A... peut être utilisée comme équivalent du questionnaire anglais YFAS 2.0 pour étudier l'addiction à la nourriture dans les populations arabophones. — Arabic-population food addiction validation study; non-French population
MORFITT-D31	Un autoquestionnaire envoyé par courrier électronique a été rempli par 191 médecins internes sur 207 travaillant dans différentes spécialités et régions de Riyad. — Saudi Arabian intern survey; non-Metropolitan French context

WEB-1	EHRNoteQA (NeurIPS 2024) demonstrates that existing clinical NLP benchmarks typically fail to represent raw multi-note EHR documentation; no equivalent French benchmark identified (https://arxiv.org/abs/2402.16040)
WEB-2	PMSI uses GHM groupings derived from CIM-10 FR principal/secondary diagnoses + CCAM procedures + age + length of stay — substantially more granular than ICD-10 chapter labels (https://www.lespmsi.com/2024/manuel_GHM_2024_volume1_provisoire.pdf)
WEB-18	Recent 2025 meta-analysis confirms a significant proportion of medical LLM benchmarks are constructed from academic materials rather than authentic clinical data — applicable to DrBenchmark's source composition (https://arxiv.org/html/2508.04325)

Information Gaps

- Quantitative measure of how performance on edited-register benchmarks transfers to noisy active EHR documentation in metropolitan French hospitals
- Full geographic distribution across all 739 DiaMed cases (only sample-confirmed as non-metropolitan)

Requires Expert Verification

- Whether French clinical abbreviation handling (ATCD, TTT, HDJ, etc.) by models trained on DrBenchmark is sufficient for deployment without a custom normalization layer

Remediation

Gap 1: No DrBenchmark dataset uses raw active-hospital EHR notes; benchmark sources are systematically cleaner than the deployment's moderately noisy hospital documentation, and DiaMed/MorFITT introduce non-metropolitan content (Burkina Faso, Niger, Morocco, Quebec, Jordan, Saudi Arabia)

Recommendation: Construct a deployment-internal validation set of metropolitan French hospital notes (with appropriate HDS-certified hosting [WEB-4]) annotated by hospital coders, sized for stratified evaluation across PMSI Catégories Majeures. Filter MorFITT and DiaMed evaluation results to exclude non-metropolitan cases when reporting deployment-relevant performance.

Gap 2: French clinical abbreviations (ATCD, TTT, HDJ, etc.) characteristic of hospital notes are under-represented in benchmark sources

Recommendation: Build a custom abbreviation expansion / normalization layer using French hospital coding documentation (CI-SIS / ANS standards) and validate it against held-out hospital notes before relying on DrBenchmark-trained models for production NER outputs.

ID	Evidence
WEB-4	https://engage.hoganlovells.com/knowledgeservices/news/health-data-hosting-the-new-french-hds-certification-has-been-released

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input Form was flagged LOWER priority in elicitation, and the benchmark aligns well on this dimension. DrBenchmark is exclusively text-based in standard written French [\[Q31\]](#), implemented via HuggingFace Datasets, Transformers, and PyTorch [\[Q48\]](#), matching the deployment’s text-only clinical NLP pipeline. The Latin alphabet with French diacritics is the natural encoding for both benchmark and deployment, with no script mismatch (QUAERO-D9, MANTRAGSC-D4). Tokenization behavior is investigated through average sub-token counts per word [\[Q75, Q89\]](#), and document splitting at sentence boundaries [\[Q37\]](#) mirrors typical preprocessing. Minor concerns: (1) DEFT-2020 contains a single Japanese-script example (DEFT2020-D13) which is an isolated artifact; (2) some PxCORPUS examples contain ASR artifacts, garbled transcriptions, and embedded system prompts that mix register and form (PXCORPUS-D3, PXCORPUS-D22, PXCORPUS-D27); (3) CLISTER includes degenerate tabular-row inputs (CLISTER-D17) and OCR/PDF extraction artifacts like mid-word line breaks (CLISTER-D18); (4) E3C contains single-punctuation tokenization artifacts in Basque configs (E3C-D15, E3C-D16). None of these compromises the fundamental form alignment for the deployment.

ID	Evidence
Q31	“While the dataset covers 5 languages, only the French portion is retained for the benchmark.” (p.3)
Q37	“Due to the presence of nested entities in annotations, we simplified the evaluation process by retaining only annotations at the higher granularity level from the BigBio (Fries et al., 2022) implementation, following the approach described in Touchent et al. (2023), which translates into an average loss of 6.06% of the annotations on EMEA and 8.90% on MEDLINE. Additionally, considering that some documents from EMEA exceed the maximum input sequence length that most current language models can handle, we decided to split these documents into sentences.” (p.3)
Q48	“We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries.” (p.5)
Q75	“Tokenizers play a crucial role in MLMs by utilizing size-limited vocabularies to split texts into sub-units, aiming to handle out-of-vocabulary (OOV) words.” (p.8)
Q89	“Firstly, we investigated the influence of tokenizers based on the average number of sub-tokens they produce per word.” (p.9)
PXCORPUS-D3	/chet — Single garbled transcription artifact labeled 'none'
MANTRAGSC-D4	Des symptômes potentiellement liés à l' histamine tels que éruption cutanée étendue, gonflement du visage et/ ou des lèvres, démangeaisons, sensation de chaleur ou difficulté à respirer, ont été rapportés. — Well-formed regulatory French with dense symptom NER; confirms formal register alignment
QUAERO-D9	Une cirrhose du foie peut également affecter l'excrétion rénale de la lépirudine. — Standard written French medical prose; confirms script/register alignment
DEFT2020-D13	42 Jeux indémodables (だれでもアソビ大全 , Daredemo Asobi Taizen au Japon...) — Japanese script embedded in French text; video game content; off-domain
E3C-D15	"." tokens=['.'] ner_tags=[0] — Single-punctuation example — tokenization artifact, quality concern
E3C-D16	"." tokens=['.'] ner_tags=[0] — Second single-punctuation example — repeated artifact
CLISTER-D17	Jour 45 - - - - - 25 µg 0,4 mg — Table row, not a clinical sentence; degenerate STS input
CLISTER-D18	Sheldon).\n\nLe patient subissait une chi miothérapie par Méthotrexate - Vinblastine - Endoxan -Cisplatine par voie — Line-break mid-word artifact ("chi miothérapie"); OCR/PDF extraction noise in source material

PXCORPUS-D 22	teralithe 250 milligrammes / le serveur de dialogue met beaucoup de temps à comprendre votre énoncé veuillez reformuler différemment teralithe 250 milligrammes 2 comprimés matin et soir pendant 1 mois — System prompt embedded in prescription utterance; corrupts register
PXCORPUS-D 27	tropatopine chlorhydrate 10 milligrammes 1 comprimé le midi — "1o" instead of "10" — ASR/OCR artifact in dose quantity

Strengths

- Exclusively text-only standard French aligned to deployment's text-only pipeline [Q31, Q48]
- Latin alphabet with French diacritics natively supported throughout (QUAERO-D9, MANTRAGSC-D4, DEFT2021-D1)
- HuggingFace/PyTorch implementation provides reproducible toolkit matching modern French biomedical NLP practice [Q48]
- Tokenization investigated empirically across 8 MLMs [Q75, Q78, Q89]

ID	Evidence
Q31	"While the dataset covers 5 languages, only the French portion is retained for the benchmark." (p.3)
Q48	"We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries." (p.5)
Q75	"Tokenizers play a crucial role in MLMs by utilizing size-limited vocabularies to split texts into sub-units, aiming to handle out-of-vocabulary (OOV) words." (p.8)
Q78	"However, our experiments, as shown in Table 7, reveal that FlauBERT is the model with the least word segmentation (1.43 in average), while DrBERT-CP tends to have the highest average segmentation (1.90 in average)." (p.8)
Q89	"Firstly, we investigated the influence of tokenizers based on the average number of sub-tokens they produce per word." (p.9)
DEFT2021-D1	Un homme de 42 ans était hospitalisé en urgence le 10 décembre 2003 dans le service de gastro-entérologie pour un tableau d'hépatite aiguë — Multi-morbid hepatology/toxicology/psychiatry case with 6 concurrent MeSH labels
MANTRAGSC-D4	Des symptômes potentiellement liés à l'histamine tels que éruption cutanée étendue, gonflement du visage et/ ou des lèvres, démangeaisons, sensation de chaleur ou difficulté à respirer, ont été rapportés. — Well-formed regulatory French with dense symptom NER; confirms formal register alignment
QUAERO-D9	Une cirrhose du foie peut également affecter l'excrétion rénale de la lépirudine. — Standard written French medical prose; confirms script/register alignment

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both benchmark and deployment are text-only, standard written French with French diacritics. No mismatch in signal encoding.

IF-2: Regional infrastructure (French hospital IT) supports the same UTF-8 / Latin-diacritical text encoding. HDS-certified hosting requirements [WEB-4] govern data hosting but do not affect signal form.

IF-3: Domain-specific form: tokenization of medical compounds and abbreviations. DrBenchmark studies sub-token counts across 8 models [Q75, Q78] and finds tokenization plays a minor role in performance [Q80].

IF-4: Minor form mismatches: DEFT-2020 contains incidental Japanese script (DEFT2020-D13); PxCorpus contains ASR artifacts and embedded system prompts (PXCORPUS-D3, PXCORPUS-D22); CLISTER contains OCR/PDF extraction noise (CLISTER-D18). These are isolated and do not threaten external validity for the deployment.

ID	Evidence
Q31	"While the dataset covers 5 languages, only the French portion is retained for the benchmark." (p.3)
Q48	"We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries." (p.5)
Q75	"Tokenizers play a crucial role in MLMs by utilizing size-limited vocabularies to split texts into sub-units, aiming to handle out-of-vocabulary (OOV) words." (p.8)
Q78	"However, our experiments, as shown in Table 7, reveal that FlauBERT is the model with the least word segmentation (1.43 in average), while DrBERT-CP tends to have the highest average segmentation (1.90 in average)." (p.8)
Q80	"Yet, tokenization, as it is currently done in MLMs, seems to play a minor role in the performance of systems." (p.8)
PXCORPUS-D3	/chet — Single garbled transcription artifact labeled 'none'
QUAERO-D9	Une cirrhose du foie peut également affecter l'excrétion rénale de la lépirudine. — Standard written French medical prose; confirms script/register alignment
DEFT2020-D13	42 Jeux indémodables (だれでもアソビ大全 , Daredemo Asobi Taizen au Japon...) — Japanese script embedded in French text; video game content; off-domain
CLISTER-D18	Sheldon).\n\nLe patient subissait une chi miothérapie par Méthotrexate - Vinblastine - Endoxan -Cisplatine par voie — Line-break mid-word artifact ("chi miothérapie"); OCR/PDF extraction noise in source material
PXCORPUS-D22	teralithe 250 milligrammes / le serveur de dialogue met beaucoup de temps à comprendre votre énoncé veuillez reformuler différemment teralithe 250 milligrammes 2 comprimés matin et soir pendant 1 mois — System prompt embedded in prescription utterance; corrupts register
WEB-4	https://engage.hoganlovells.com/knowledgeservices/news/health-data-hosting-the-new-french-hds-certification-has-been-released

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output Ontology was flagged HIGH priority in elicitation due to the structural mismatch between the deployment's confidence-scored multi-label outputs with uncertainty flagging and the benchmark's standard classification metrics. The benchmark provides rich label ontologies — POS tags (CAS 31, ESSAI 36) [Q101, Q102], NER schemes (QUAERO 10 UMLS Semantic Groups [Q103, Q35], E3C clinical/temporal [Q104], DEFT-2021 14 entity types [Q108], Mantra-GSC [Q106]), DEFT-2021 23 MeSH axes [Q27], DiaMed 22 ICD-10 chapters [Q45, Q107], MorFITT 12 specialties [Q105]. DEFT-2021's NER scheme covers exactly the entity types the deployment needs (PATHOLOGY, ANATOMY, TREATMENT, DOSAGE, DURATION, FREQUENCY, MOMENT, DATE) [Q108, DEFT2021-D3, DEFT2021-D4, DEFT2021-D5]. However, multiple structural mismatches harm output ontology validity: (1) CRITICAL — no benchmark task has confidence-scored, ranked-candidate, or uncertainty-flagging outputs; DEFT-2021's specialties_one_hot is strictly binary 0/1 (DEFT2021-D11), DiaMed is single-label per case (DIAMED-D10), and DEFT-2020 averages annotator scores into a single float (DEFT2020-D16); (2) ICD-10 chapter granularity in DiaMed (22 categories) is far coarser than operational PMSI/GHM codes derived from CIM-10 FR (DIAMED-D11) [WEB-2]; (3) several datasets have output taxonomies orthogonal to the deployment's pathology classification function — CAS/ESSAI measure negation/speculation modality (CAS-D19, ESSAI-D9), FrenchMedMCQA measures answer-set selection (FRENCHMEDMCQA-D1), DEFT-2020 measures sentence similarity (DEFT2020-D14, DEFT2020-D15); (4) UMLS Semantic Group NER provides no ICD-10 sub-typing (MANTRAGSC-D10); (5) E3C French_clinical uses only binary CLINENTITY annotation, underspecifying multi-type pathology distinction (E3C-D14); (6) QUAERO and Mantra-GSC contain no temporal entity labels at all (QUAERO-D13, QUAERO-D14, MANTRAGSC-D9); (7) PxCorpus NER lacks pathology/diagnosis entities (PXCORPUS-D16). The benchmark explicitly notes that even specialized models perform poorly on MLC and STS [Q82, Q86, Q87], directly implicating the deployment's most challenging function.

ID	Evidence
Q27	"The dataset is annotated with 23 axes from Chapter C of the Medical Subject Headings (MeSH)." (p.3)
Q35	"10 entity categories corresponding to the UMLS Semantic Groups (Lindberg et al., 1993) were annotated (more detail in Appendix B.3)." (p.3)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
Q82	"Upon analyzing the average performance by task category, it becomes evident that the leading model, DrBERT-FS, does not excel in tasks such as MLC or STS." (p.8)
Q86	"We have observed that several biomedical tasks in DrBenchmark exhibit relatively poor performance, even when utilizing specialized biomedical models." (p.9)
Q87	"We postulate that the models examined in this study, here state-of-the-art MLMs, may not be the most effective choices for specific tasks such as question-answering or multi-label classification." (p.9)
Q101	"INT, PRO:DEM, VER:impf, VER:ppre, PRP:det, KON, VER:ppe, PRP, PRO:IND, VER:simp, VER:con, SENT, VER:futu, PRO:PER, VER:infi, ADJ, NAM, NUM, PUN:cit, PRO:REL, VER:subi, ABR, NOM, VER:pres, DET:ART, VER:cond, VER:subp, DET:POS, ADV, SYM and PUN." (p.14)

Q102	“INT, PRO:POS, PRP, SENT, PRO, ABR, VER:pres, KON, SYM, DET:POS, VER:, PRO:IND, NAM, ADV, PRO:DEM, NN, PRO:PER, VER:pper, VER:ppre, PUN, VER:simp, PREF, NUM, VER:futu, NOM, VER:impf, VER:subp, VER:infi, DET:ART, PUN:cit, ADJ, PRP:det, PRO:REL, VER:cond and VER:subi.” (p.14)
Q103	“O, GEOG, PHEN, DISO, ANAT, OBJC, PHYS, PROC, DEVI, CHEM and LIVB” (p.14)
Q104	“Clinical: O, and CLINENTITY. Temporal: O, EVENT, ACTOR, BODYPART, TIMEX3 and RML” (p.14)
Q105	“microbiology, etiology, virology, physiology, immunology, parasitology, genetics, chemistry, veterinary, surgery, pharmacology and psychology” (p.14)
Q106	“Medline: ANAT, PROC, CHEM, PHYS, GEOG, DEVI, LIVB, OBJC, DISO, PHEN and O. EMEA and Patents: ANAT, PROC, CHEM, PHYS, DEVI, LIVB, OBJC, DISO, PHEN and O.” (p.14)
Q107	“Multi-label Classification: immunitaire (immunology), endocriniennes (endocrinology), blessures (injury), chimiques (chemicals), etatsosy (signs and symptoms), nutritionnelles (nutrition), infections (infections), virales (virology), parasitaires (parasitology), tumeur (oncology), osteomusculaires (osteomuscular disorders), stomatognathique (stomatology), digestif (digestive system disorders), respiratoire (respiratory system disorders), ORL (otorhinolaryngologic diseases), nerveux (nervous system disorders), oeil (eye diseases), homme (male genital diseases), femme (female genital diseases), cardiovasculaires (cardiology), hemopathies (hemic and lymphatic diseases), genetique (genetic disorders) and peau (dermatology).” (p.14)
Q108	“Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE” (p.14)
FRENCHMED MCQA-D1	Au cours de la leucémie lymphoïde chronique, le myélogramme montre: Une population de lymphocytes>30% — Clinical hematology question in formal medical French
DEFT2021-D3	cotrimoxazole pour 10 jours — Treatment drug with duration span annotated — confirms temporal treatment labeling
DEFT2021-D4	Hydrocortisone 300 mg IV immédiatement ; Traitement Hydroxyzine 25 mg par voie orale toutes les 6 heures — Multiple entity types co-occurring in a single sentence, confirming NER scheme coverage
DEFT2021-D5	Au jour 42 , une deuxième désensibilisation a eu lieu — Temporal anchor (relative day) labeled, confirming temporal entity extraction alignment
ESSAI-D9	Essai clinique d'immunothérapie évaluant l'association du galunisertib (un facteur de croissance transformant β), avec l'anticorps anti-PD-L1 durvalumab (MEDI4736), dans le cancer du pancréas métastatique — Rich pathology content labeled only for negation/speculation, not ICD-10 — output ontology mismatch
MANTRAGSC-D9	Traitement En cas d' ingestion récente, l' éventualité de provoquer un vomissement et d' effectuer un lavage gastrique devra être considérée. — Temporal word "récente" present but no temporal NER label exists in Mantra-GSC scheme
DIAMED-D10	Le traitement était constitué d'énoxaparine d'antiagrégant plaquettaire, de diurétique de l'anse, de dérivés morphiniques et d'oxygène — Multi-system case forced into single digestive label — masks multi-morbidity
MANTRAGSC-D10	Nouveautés dans la dysplasie ventriculaire droite arythmogène. — Single DISO class applied to cardiovascular disorder; no ICD-10 subcategory distinction
DEFT2021-D11	[1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 1.0, 1.0, 0.0, 1.0] — Strictly binary 0/1 label vector — no confidence or soft-label dimension present
DIAMED-D11	Maladie de Kaposi à localisation broncho-pulmonaire révélant une infection VIH — Chapter-level label only; PMSI granular codes (e.g., B21.x) not evaluated
QUAERO-D13	La durée du traitement est allée d'une heure de perfusion en bolus à une utilisation continue de plus de 6 ans. — Temporal expressions present but unlabeled; confirms absence of temporal NER
DEFT2020-D14	Fields: id, moy, vote, scores, source, cible — Continuous similarity scoring schema; no entity labels, no ICD-10 classification outputs
E3C-D14	"acute respiratory distress syndrome" tagged as single CLINENTITY span, no sub-type — Coarse two-class annotation lacks pathology/anatomy/treatment distinction needed by deployment
QUAERO-D14	La dose peut être augmentée par intervalles de 1 à 2 jours, voire plus... — Another unlabeled temporal phrase; reinforces temporal NER gap
DEFT2020-D15	Field: correct_cible (0/1/2) — Sentence selection index; not a pathology classification or ICD-10 label
DEFT2020-D16	Après inspection, elles ont toutes été exclues. — Annotator disagreement averaged away in scoring; calibration signal lost
PXCORPUS-D16	NER tags: dose, form, frequency, duration, symptom only — No PATHOLOGY/DIAGNOSIS entity in tag scheme; ICD-10 mapping untestable
CAS-D19	label_names: ["negation_speculation", "negation", "neutral", "speculation"] — Classification labels measure linguistic modality, not ICD-10 pathology categories
WEB-2	PMSI uses GHM groupings derived from CIM-10 FR principal/secondary diagnoses + CCAM procedures + age + length of stay — substantially more granular than ICD-10 chapter labels (https://www.lespmsi.com/2024/manuel_GHM_2024_volume1_provisoire.pdf)

Strengths

- DEFT-2021 NER ontology covers all entity types the deployment requires: PATHOLOGY, ANATOMY, TREATMENT, DOSAGE, DURATION, FREQUENCY, MOMENT, DATE [Q108, DEFT2021-D3, DEFT2021-D4]
- DiaMed provides ICD-10 chapter-level labels aligned with the deployment's classification ontology, the only direct ICD-10 alignment in the benchmark (DIAMED-D1, DIAMED-D2)
- DEFT-2021 23 MeSH axes support multi-label structure with up to 7 concurrent labels per case (DEFT2021-D1, DEFT2021-D2)
- Benchmark's own findings flag MLC/STS as poor-performance areas [Q82, Q86] — useful diagnostic signal for deployment risk

ID	Evidence
Q82	"Upon analyzing the average performance by task category, it becomes evident that the leading model, DrBERT-FS, does not excel in tasks such as MLC or STS." (p.8)
Q86	"We have observed that several biomedical tasks in DrBenchmark exhibit relatively poor performance, even when utilizing specialized biomedical models." (p.9)
Q108	"Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE" (p.14)
DIAMED-D1	Le test rapide VIH était positif, confirmé par la sérologie VIH avec un taux de CD4 à 27/mm3 — Confirmed HIV infection with CD4 count — supports infectious disease label
DEFT2021-D1	Un homme de 42 ans était hospitalisé en urgence le 10 décembre 2003 dans le service de gastro-entérologie pour un tableau d'hépatite aiguë — Multi-morbid hepatology/toxicology/psychiatry case with 6 concurrent MeSH labels
DIAMED-D2	L'examen anatomopathologique de la pièce opératoire montrait une prolifération tumorale faite de travées, de cordons — Histopathological finding of tumoral proliferation — confirms neoplasm label
DEFT2021-D2	Nous rapportons le cas d'un garçon de huit ans [...] microangiopathie thrombotique secondaire à un syndrome hémolytique urémique (SHU) atypique — 7-label pediatric multi-morbid case illustrating complex label overlap
DEFT2021-D3	cotrimoxazole pour 10 jours — Treatment drug with duration span annotated — confirms temporal treatment labeling
DEFT2021-D4	Hydrocortisone 300 mg IV immédiatement ; Traitement Hydroxyzine 25 mg par voie orale toutes les 6 heures — Multiple entity types co-occurring in a single sentence, confirming NER scheme coverage

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories partially align: NER tag schemes (DEFT-2021, E3C temporal) cover deployment-needed entity types [Q108, Q104]; ICD-10 chapter labels in DiaMed align coarsely with PMSI mapping [Q45]; POS tag inventories match standard French biomedical morphology [Q101, Q102].

OO-2: Missing categories: (1) confidence scores / ranked candidates / uncertainty flags — absent across all 20 tasks [Q56, DEFT2021-D11, DIAMED-D10]; (2) PMSI-granular codes — DiaMed only at chapter level (DIAMED-D11) [WEB-2]; (3) tropical/DOM-TOM-specific pathology subcategories — absent from French subsets (E3C-D4, E3C-D13); (4) pathology entities in PxCorpus NER (PXCORPUS-D16); (5) temporal labels in QUAERO and Mantra-GSC (QUAERO-D13, MANTRAGSC-D9).

OO-3: MorFITT's 12 medical specialty labels (microbiology, surgery, pharmacology) encode research-specialty routing rather than ICD-10 diagnostic axes, and do not reflect French hospital coding conventions (MORFITT-D11, MORFITT-D21). CAS/ESSAI's negation/speculation modality measures linguistic phenomena, not pathology classification (CAS-D19, ESSAI-D9). DEFT-2020 STS continuous similarity score has no diagnostic correspondence (DEFT2020-D14).

OO-4: A stakeholder-driven taxonomy redesign would be needed to introduce: (a) calibrated confidence scoring layer over multi-label outputs; (b) PMSI/GHM code-level granularity beyond ICD-10 chapter; (c) ranked-candidate output format with uncertainty thresholds. The deployment's Silver Standard framing requires these for clinician adjudication.

OO-5: Structural validity is harmed by absence of confidence/uncertainty output structure [Q56, DEFT2021-D11]. Content validity is harmed by orthogonal classification labels (negation/speculation in CAS/ESSAI; STS continuous in DEFT-2020/CLISTER; pharmacy MCQA in FrenchMedMCQA) and by chapter-level rather than PMSI-level granularity [WEB-2]. The benchmark itself observes that 'biomedical tasks in DrBenchmark exhibit relatively poor performance, even when utilizing specialized biomedical models' [Q86] and 'state-of-the-art MLMs may not be the most effective choices for specific tasks such as question-answering or multi-label classification' [Q87].

ID	Evidence
Q27	"The dataset is annotated with 23 axes from Chapter C of the Medical Subject Headings (MeSH)." (p.3)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
Q56	"The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions." (p.6)
Q86	"We have observed that several biomedical tasks in DrBenchmark exhibit relatively poor performance, even when utilizing specialized biomedical models." (p.9)
Q87	"We postulate that the models examined in this study, here state-of-the-art MLMs, may not be the most effective choices for specific tasks such as question-answering or multi-label classification." (p.9)
Q101	"INT, PRO:DEM, VER:impf, VER:ppre, PRP:det, KON, VER:pper, PRP, PRO:IND, VER:simp, VER:con, SENT, VER:futu, PRO:PER, VER:infi, ADJ, NAM, NUM, PUN:cit, PRO:REL, VER:subi, ABR, NOM, VER:pres, DET:ART, VER:cond, VER:subp, DET:POS, ADV, SYM and PUN." (p.14)
Q102	"INT, PRO:POS, PRP, SENT, PRO, ABR, VER:pres, KON, SYM, DET:POS, VER:, PRO:IND, NAM, ADV, PRO:DEM, NN, PRO:PER, VER:pper, VER:ppre, PUN, VER:simp, PREF, NUM, VER:futu, NOM, VER:impf, VER:subp, VER:infi, DET:ART, PUN:cit, ADJ, PRP:det, PRO:REL, VER:cond and VER:subi." (p.14)
Q104	"Clinical: O, and CLINENTITY. Temporal: O, EVENT, ACTOR, BODYPART, TIMEX3 and RML" (p.14)
Q108	"Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE" (p.14)
DEFT2021-D3	cotrimoxazole pour 10 jours — Treatment drug with duration span annotated — confirms temporal treatment labeling
E3C-D4	The presence of terminal spine confirmed Schistosoma haematobium and a diagnosis of schistosomiasis was made. — Tropical parasitic pathology in English config — confirms tropical content structurally absent from French subset
MANTRAGSC-D9	Traitement En cas d' ingestion récente, l' éventualité de provoquer un vomissement et d' effectuer un lavage gastrique devra être considérée. — Temporal word "récente" present but no temporal NER label exists in Mantra-GSC scheme
ESSAI-D9	Essai clinique d'immunothérapie évaluant l'association du galunisertib (un facteur de croissance transformant β), avec l'anticorps anti-PD-L1 durvalumab (MEDI4736), dans le cancer du pancréas métastatique — Rich pathology content labeled only for negation/speculation, not ICD-10 — output ontology mismatch
DIAMED-D10	Le traitement était constitué d'énoxaparine d'antiagrégant plaquettaire, de diurétique de l'anse, de dérivés morphiniques et d'oxygène — Multi-system case forced into single digestive label — masks multi-morbidity
DEFT2021-D11	[1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 1.0, 1.0, 0.0, 1.0] — Strictly binary 0/1 label vector — no confidence or soft-label dimension present
DIAMED-D11	Maladie de Kaposi à localisation broncho-pulmonaire révélant une infection VIH — Chapter-level label only; PMSI granular codes (e.g., B21.x) not evaluated
MORFITT-D11	Nous rapportons le cas d'une patiente de 71 ans ayant développé un pseudomyxome péritonéal secondaire à un adénocarcinome appendiculaire. — Clinical case abstract with surgery+etiology labels; labels do not map to ICD-10 codes

QUAERO-D13	La durée du traitement est allée d'une heure de perfusion en bolus à une utilisation continue de plus de 6 ans. — Temporal expressions present but unlabeled; confirms absence of temporal NER
E3C-D13	a diagnosis of schistosomiasis was made — Tropical disease in English config only; absent from French subset — confirms DOM-TOM pathology gap
DEFT2020-D14	Fields: id, moy, vote, scores, source, cible — Continuous similarity scoring schema; no entity labels, no ICD-10 classification outputs
PXCORPUS-D16	NER tags: dose, form, frequency, duration, symptom only — No PATHOLOGY/DIAGNOSIS entity in tag scheme; ICD-10 mapping untestable
CAS-D19	label_names: ["negation_speculation", "negation", "neutral", "speculation"] — Classification labels measure linguistic modality, not ICD-10 pathology categories
MORFITT-D21	Bloquer le complément, notamment l'axe C5a-C5aR1, par des thérapies spécifiques représente un espoir thérapeutique dans les formes les plus sévères de la maladie. — COVID-19 immunology; three specialty labels that don't resolve to ICD-10 output
WEB-2	PMSI uses GHM groupings derived from CIM-10 FR principal/secondary diagnoses + CCAM procedures + age + length of stay — substantially more granular than ICD-10 chapter labels (https://www.lespmsi.com/2024/manuel_GHM_2024_volume1_provisoire.pdf)

Information Gaps

- Whether the 23 MeSH axes in DEFT-2021 map systematically to PMSI Catégories Majeures groupings

Requires Expert Verification

- Calibration thresholds appropriate for the deployment's uncertainty-flagging behavior; benchmark provides no protocol for setting these

Remediation

Gap 1: No benchmark task evaluates confidence-scored multi-label outputs, uncertainty flagging, or ranked-candidate quality — the deployment's central output behavior is structurally unmeasurable; ICD-10 chapter granularity is coarser than operational PMSI/GHM codes

Recommendation: Extend the DrBenchmark protocol on DEFT-2021 and DiaMed with calibration metrics (Expected Calibration Error, Brier score, per-label AUROC, top-k ranked-candidate F1). Build a supplementary PMSI-code-level evaluation set using a hospital-internal stratified sample mapped to CIM-10 FR / GHM groupings per ATIH 2024 guidance [WEB-2].

ID	Evidence
WEB-2	PMSI uses GHM groupings derived from CIM-10 FR principal/secondary diagnoses + CCAM procedures + age + length of stay — substantially more granular than ICD-10 chapter labels (https://www.lespmsi.com/2024/manuel_GHM_2024_volume1_provisoire.pdf)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output Content was flagged HIGH priority due to the deployment's Silver Standard framing and the explicit gap between academic NLP annotators and practicing PMSI coders/hospital clinicians. Findings strongly confirm this concern. Annotation provenance varies dramatically across the 20 datasets and is generally undocumented for clinical-coding authority. DiaMed is the only dataset with explicit IAA statistics and medical expert involvement (one expert) [\[Q45, Q46\]](#); CAS POS tags are silver-standard automatic via Tagex [\[Q42, CAS-D15\]](#); ESSAI POS tags are automatic via TreeTagger [\[Q43\]](#); DEFT-2021's MeSH multi-label and NER annotations are described as 'manually annotated' but annotator credentials are not documented [\[Q25, DEFT2021-D16\]](#); CLISTER STS scores are averaged across multiple annotators of unspecified expertise [\[Q41, CLISTER-D12\]](#). Web-source confirmation: no PMSI coder or hospital coding specialist involvement is documented in any DrBenchmark constituent dataset. Multiple dataset-level findings show this matters: CLISTER-D12 ('lost to follow-up' vs. 'death at J13' rated as 1.0 similarity — clinically critical distinction underweighted by non-specialist annotators); CLISTER-D13 (clinical abbreviation 'UB' likely Buruli ulcer requires domain expertise); E3C-D11, E3C-D12 (identifiable pathogens like *Nocardia asteroides* and treatments left unlabeled — annotation incompleteness); CAS-D17 (annotation rationale opaque for ambiguous speculation labels); ESSAI-D14 (consecutive all-zero NER sequences despite clinically relevant content). DiaMed's annotation is further compromised for the metropolitan deployment by sub-Saharan/North African case origin (DIAMED-D7, DIAMED-D8, DIAMED-D9) — the labels reflect Pan African Medical Journal cases, not metropolitan French hospital coding. Author affiliations are exclusively academic (Nantes Université, Avignon Université, INSERM, CNRS, Université de Lille) [\[Q7, Q8\]](#) with no PMSI coder or hospital coding specialist representation. The deployment's explicit annotator-vs-user gap concern is fully confirmed.

ID	Evidence
Q7	"Yanis Labrak, Adrien Bazoge, Oumaima El Khetari, Mickael Rouvier, Pacôme Constant dit Beaufils, Natalia Grabar, Béatrice Daille, Solen Quiniou, Emmanuel Morin, Pierre-Antoine Gourraud and Richard Dufour" (p.1)
Q8	"LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille" (p.1)
Q25	"This dataset is manually annotated in two tasks: (i) multi-label classification and (ii) NER." (p.3)
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
Q42	"CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)
Q43	"ESSAI (Dalloux et al., 2021) contains 7,247 clinical trial protocols annotated in 41 POS tags using TreeTagger (Schmid, 1994). As the dataset was not originally divided into 3 subsets, we applied the same procedure as on the CAS corpus." (p.4)

Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
Q46	"Table 4: Inter-annotator agreement statistics. κ is referring to Kappa Cohen and G to Gwet's AC1." (p.4)
DIAMED-D7	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabe hospital origin — non-metropolitan French clinical setting
DIAMED-D8	À propos d'un cas clinique ... keywords: ['Niamey'] — Niger-based case — sub-Saharan African clinical context, not metropolitan France
DIAMED-D9	keywords: ['Rate', 'abdomen aigu', 'scanner abdominal', 'chirurgie', 'Maroc'] — Moroccan clinical case — North African origin rather than French metropolitan
E3C-D11	"La culture a isolé le germe Nocardia asteroides." all tags [0,0,0,0,0,0,0,0] — "Nocardia asteroides" (pathogen) not tagged as CLINENTITY — suggests annotation incompleteness
CLISTER-D12	Le patient a été perdu de vue." vs. "Le patient décède à J13. — Outcome divergence (lost to follow-up vs. death on day 13); clinically significant difference potentially underweighted by non-specialist annotators
E3C-D12	"Le patient a été mis sous traitement par ciclosporine avec une évolution rapide vers une leucémie aigue myéloblastique." all tags [0,...,0] — Clear pathology and treatment entities untagged — annotation completeness concern
CLISTER-D13	Un patient de 42 ans, a été hospitalisé pour un UB de la main droite. — Clinical abbreviation "UB" (likely Buruli ulcer); domain expertise needed for reliable similarity annotation
ESSAI-D14	"avec la combinaison gemcitabine + abraxane, chez des patients avec un cancer du pancréas" (and 9 similar) — Consecutive all-zero NER sequences despite clinically relevant content — sparse annotation density
CAS-D15	annotated for POS tagging with 31 classes using automatic annotations through Tagex 3...yielded 98% precision — Silver-standard label provenance for POS config
DEFT2021-D16	(see D6 excerpt) — Multi-drug cardiac case; annotator PMSI credential unknown — label authority concern
CAS-D17	"Cholstat ® 0.1 ." — Drug name/dosage fragment labeled as speculation with no apparent epistemic marker; annotation rationale opaque

Strengths

- DiaMed provides documented IAA with Cohen's Kappa and Gwet's AC1 across two annotation sessions and explicit medical-expert involvement [Q45, Q46]
- DEFT-2021 is described as manually annotated for both MLC and NER tasks [Q25]
- CAS Tagex automatic POS annotations were validated against manual annotations at 98% precision [Q42] — partial quality assurance
- Benchmark authors include clinical-affiliated institutions (CHU Nantes, Service de Neuroradiologie, l'institut du thorax) [Q8] indicating clinical proximity even if not PMSI-coder authority

ID	Evidence
Q8	"LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille" (p.1)
Q25	"This dataset is manually annotated in two tasks: (i) multi-label classification and (ii) NER." (p.3)
Q42	"CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)

Q46	"Table 4: Inter-annotator agreement statistics. κ is referring to Kappa Cohen and G to Gwet's AC1." (p.4)
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Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels do not reflect practicing PMSI coder or hospital clinical coding specialist perspectives. The single documented medical expert is in DiaMed [Q45], whose cases are non-metropolitan (DIAMED-D7, DIAMED-D8). Labels reflect academic NLP annotator judgment, not the operational hospital coding population whose authority the deployer requires.

OC-2: Significant disagreement potential. Examples: CLISTER-D12 (clinically distinct outcomes 'lost to follow-up' vs. 'death' rated as low similarity but clinical significance is asymmetric); CLISTER-D13 ('UB' abbreviation requires domain expertise non-specialist annotators may lack); E3C-D11, E3C-D12 (pathogens and treatments left unlabeled — practicing clinicians would likely tag these). Documented PMSI coding quality concerns [WEB-19, WEB-20] independently confirm that even within practicing coder populations, ICD coding has documented inter-coder variability.

OC-3: Annotator demographics are minimally documented. Author affiliations [Q7, Q8] are academic NLP and clinical research institutions; no Datasheet or Data Statement at annotator level is published for most constituent datasets. CAS uses automatic Tagex annotations [Q42, CAS-D15]. ESSAI uses automatic TreeTagger [Q43]. DEFT-2021 MeSH annotation provenance is described only as 'manually annotated' [Q25, DEFT2021-D16]. CLISTER documents 'several annotators' without expertise breakdown [Q41, CLISTER-D12].

OC-4: Re-annotation by a representative regional pool (PMSI coders + hospital clinicians) is recommended for at least DEFT-2021 (the deployment's most aligned multi-label task) and DiaMed (the only ICD-10-labeled task). The deployment's Silver Standard framing implicitly already plans for clinician adjudication; the benchmark itself does not provide labels validated by this population.

OC-5: Aggregation methods erase minority perspectives. CLISTER averages multi-annotator similarity into a single float [Q41, DEFT2020-D16]; CAS and ESSAI use silver-standard automatic POS tags with no human disagreement signal [CAS-D15]; DiaMed's IAA captures inter-annotator agreement but final labels are not multi-label or confidence-scored (DIAMED-D10). The aggregation discards the very disagreement signal that could inform deployment uncertainty flagging.

OC-6: Convergent validity is harmed: labels do not correlate with PMSI coder or hospital clinician judgments in documented form. External validity is harmed: judgments by academic NLP annotators on edited published cases are not expected to generalize to active hospital coding by certified coders. Documented PMSI coding quality concerns [WEB-19] reinforce that even mapping to practicing-coder-equivalent labels is non-trivial.

ID	Evidence
Q7	"Yanis Labrak, Adrien Bazoge, Oumaima El Khetari, Mickael Rouvier, Pacôme Constant dit Beauflis, Natalia Grabar, Béatrice Daille, Solen Quiniou, Emmanuel Morin, Pierre-Antoine Gourraud and Richard Dufour" (p.1)
Q8	"LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille" (p.1)
Q25	"This dataset is manually annotated in two tasks: (i) multi-label classification and (ii) NER." (p.3)

Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
Q42	"CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)
Q43	"ESSAI (Dalloux et al., 2021) contains 7,247 clinical trial protocols annotated in 41 POS tags using TreeTagger (Schmid, 1994). As the dataset was not originally divided into 3 subsets, we applied the same procedure as on the CAS corpus." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
DIAMED-D7	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabe hospital origin — non-metropolitan French clinical setting
DIAMED-D8	À propos d'un cas clinique ... keywords: ['Niamey'] — Niger-based case — sub-Saharan African clinical context, not metropolitan France
DIAMED-D10	Le traitement était constitué d'énoxaparine d'antiagrégant plaquettaire, de diurétique de l'anse, de dérivés morphiniques et d'oxygène — Multi-system case forced into single digestive label — masks multi-morbidity
E3C-D11	"La culture a isolé le germe Nocardia asteroides." all tags [0,0,0,0,0,0,0,0] — "Nocardia asteroides" (pathogen) not tagged as CLINENTITY — suggests annotation incompleteness
CLISTER-D12	Le patient a été perdu de vue." vs. "Le patient décède à J13. — Outcome divergence (lost to follow-up vs. death on day 13); clinically significant difference potentially underweighted by non-specialist annotators
E3C-D12	"Le patient a été mis sous traitement par ciclosporine avec une évolution rapide vers une leucémie aigue myéloblastique." all tags [0,...,0] — Clear pathology and treatment entities untagged — annotation completeness concern
CLISTER-D13	Un patient de 42 ans, a été hospitalisé pour un UB de la main droite. — Clinical abbreviation "UB" (likely Buruli ulcer); domain expertise needed for reliable similarity annotation
CAS-D15	annotated for POS tagging with 31 classes using automatic annotations through Tagex 3...yielded 98% precision — Silver-standard label provenance for POS config
DEFT2021-D16	(see D6 excerpt) — Multi-drug cardiac case; annotator PMSI credential unknown — label authority concern
DEFT2020-D16	Après inspection, elles ont toutes été exclues. — Annotator disagreement averaged away in scoring; calibration signal lost
WEB-19	PMSI-MCO bacterial resistance study: ICD coding quality is unequal and some disease categories cannot be reliably characterized from PMSI coding alone, partly due to initial coding errors (https://pubmed.ncbi.nlm.nih.gov/29673880/)
WEB-20	French PMSI bacterial resistance study (ScienceDirect 2018): documented inconsistencies between infection types, causative agents, and resistance codes attributed to initial coding errors (https://www.sciencedirect.com/science/article/abs/pii/S0399077X17307072)

Information Gaps

- Specific clinical credentials of the 'medical expert' in DiaMed annotation [Q45]
- DEFT-2021 competition annotator demographic breakdown not published in benchmark documentation

Requires Expert Verification

- Empirical inter-annotator agreement between DrBenchmark labels and a sample of PMSI-coder re-annotations on a representative metropolitan French hospital subset

Remediation

Gap 1: No DrBenchmark dataset documents PMSI coder or hospital coding specialist annotation; only DiaMed has one medical expert; CAS/ESSAI POS tags are silver-standard automatic; CLISTER and E3C show

domain-expertise concerns and annotation incompleteness

Recommendation: Recruit certified PMSI coders to re-annotate a representative subset of DEFT-2021 multi-label cases and DiaMed ICD-10 assignments. Compute Cohen's Kappa / Gwet's AC1 between PMSI-coder labels and benchmark labels to quantify the annotation-authority gap explicitly. Use the disagreement signal as a calibration target for the deployment's uncertainty-flagging behavior.

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output Form was flagged MODERATE priority. The benchmark uses well-defined, reproducible scoring protocols: SeqEval with IOB2 format and first-token prediction for sequence labeling [Q52, Q53], Spearman correlation and EDRM for STS [Q54], standard classification metrics with majority-class baselines [Q56], 4-run averaging with Student's t-test [Q51], and full hyperparameter documentation [Q49, Q99, Q100]. These are well-matched to typical NER/POS/MLC pipeline outputs (E3C-D5, E3C-D6 confirm IOB2 alignment). However, the deployment requires three output forms the benchmark does not measure: (1) confidence-scored multi-label outputs, (2) uncertainty flagging for high-uncertainty entities, (3) ranked candidate lists for multi-morbid case adjudication. None of these is directly evaluable within the benchmark's scoring framework — DEFT-2021 uses binary 0/1 one-hot label vectors (DEFT2021-D11), DiaMed uses single-label per case (DIAMED-D10), and STS averages annotator scores discarding disagreement signal (DEFT2020-D16, CLISTER-D8). Several datasets exhibit severe class imbalance that masks minority-class performance under standard metrics: PxCORPUS medical_prescription ~89% vs replace <1% and negate ~2% (PXCORPUS-D14, PXCORPUS-D15); CAS cls neutral ~74% (CAS-D21); ESSAI cls neutral ~77% (ESSAI-D13). For prescription correction intents and clinical negation/speculation distinctions, standard accuracy will hide deployment-relevant failure modes. The benchmark's finding that 'all the models based on CamemBERT face difficulties in corpora with limited amount of data, such as MantraGSC Patents, where they fail to generate labels other than 'O' [Q73] is itself a manifestation of how standard metrics interact with imbalance. The output form is structurally aligned for sequence-labeling and standard classification but does not measure the calibration/uncertainty/ranking behavior central to the deployment.

ID	Evidence
Q49	"For further guidance, we have integrated all the training details, including hyperparameters, in Appendix A. This information will help users to reproduce and customize the experiments conducted with DrBenchmark." (p.5)
Q51	"All the models are fine-tuned regarding a strict protocol using the same hyperparameters for each downstream task. The reported results are obtained by averaging the scores from four separate runs, thus ensuring robustness and reliability. We also report statistical significance computed using Student's t-test." (p.5)
Q52	"To ensure a fair and consistent comparison among systems for sequence-to-sequence tasks such as POS tagging and NER, we chose the SeqEval (Nakayama, 2018) metric in conjunction with the IOB2 format and the training of all the models to predict only the label on the first token of each word as mentioned by Touchent et al. (2023)." (p.6)
Q53	"It provides a tokenizer-agnostic evaluation and mitigates any correlation between models' performances and the tokenization process." (p.6)
Q54	"For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020)." (p.6)
Q56	"The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions." (p.6)
Q73	"However, all the models based on CamemBERT face difficulties in corpora with limited amount of data, such as MantraGSC Patents, where they fail to generate labels other than 'O'." (p.7)
Q99	"For the experiments, we utilize the following hyperparameters that yield optimal performance from the models." (p.13)
Q100	"To mitigate overfitting, we locally save the best model based on its validation metric." (p.13)
E3C-D5	"Cet aspect évoquait une tumeur solide du péritoine." tokens tagged [0,0,0,0,1,2,0,0,0] — Multi-token pathology span ("tumeur solide") tagged in IOB2 — confirms NER output form alignment
E3C-D6	"As she suffered from acute respiratory distress syndrome and required mechanical ventilation..." tags [0,0,0,0,1,2,2,2,...] — Multi-token disease span annotation in B/I scheme — confirms annotation consistency

CLISTER-D8	Le reste de l'examen échographique ne trouvait aucune autre anomalie." vs. "Le reste de l'examen somatique était sans anomalie. — Intermediate similarity; different exam modality with shared negative finding structure
DIAMED-D10	Le traitement était constitué d'énoxaparine d'antiagrégant plaquettaire, de diurétique de l'anse, de dérivés morphiniques et d'oxygène — Multi-system case forced into single digestive label — masks multi-morbidity
DEFT2021-D11	[1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 1.0, 0.0, 1.0] — Strictly binary 0/1 label vector — no confidence or soft-label dimension present
ESSAI-D13	Chaque cycle dure 28 jours — Trivially neutral sentence — inflates neutral class count, contributing to severe label imbalance
PXCORPUS-D14	c'est 2 pourcent le collyre — Surface reads like prescription; labeled negate — ambiguity in minority class
PXCORPUS-D15	negate=11, replace=3 in 500 sampled — Severe class imbalance; majority-class baseline masks minority-class performance
DEFT2020-D16	Après inspection, elles ont toutes été exclues. — Annotator disagreement averaged away in scoring; calibration signal lost
CAS-D21	buffer: negation_speculation:6, negation:100, neutral:368, speculation:26 — Heavy class imbalance; neutral dominates at ~74%

Strengths

- Reproducible unified protocol: 4-run averaging, Student's t-test significance, hyperparameters in appendix [Q51, Q99, Q100]
- Tokenizer-agnostic SeqEval IOB2 evaluation aligns with standard NER pipeline output formats [Q52, Q53, E3C-D5, E3C-D6]
- MIT-licensed code release supports deployment-side audit and reproducibility [Q95]
- Explicit majority-class baselines provide a floor for classification performance interpretation [Q56]

ID	Evidence
Q51	"All the models are fine-tuned regarding a strict protocol using the same hyperparameters for each downstream task. The reported results are obtained by averaging the scores from four separate runs, thus ensuring robustness and reliability. We also report statistical significance computed using Student's t-test." (p.5)
Q52	"To ensure a fair and consistent comparison among systems for sequence-to-sequence tasks such as POS tagging and NER, we chose the SeqEval (Nakayama, 2018) metric in conjunction with the IOB2 format and the training of all the models to predict only the label on the first token of each word as mentioned by Touchent et al. (2023)." (p.6)
Q53	"It provides a tokenizer-agnostic evaluation and mitigates any correlation between models' performances and the tokenization process." (p.6)
Q56	"The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions." (p.6)
Q95	"All code for DrBenchmark is released under the MIT License." (p.9)
Q99	"For the experiments, we utilize the following hyperparameters that yield optimal performance from the models." (p.13)
Q100	"To mitigate overfitting, we locally save the best model based on its validation metric." (p.13)
E3C-D5	"Cet aspect évoquait une tumeur solide du péritoine." tokens tagged [0,0,0,0,1,2,0,0,0] — Multi-token pathology span ("tumeur solide") tagged in IOB2 — confirms NER output form alignment
E3C-D6	"As she suffered from acute respiratory distress syndrome and required mechanical ventilation..." tags [0,0,0,0,1,2,2,2,...] — Multi-token disease span annotation in B/I scheme — confirms annotation consistency

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality is text labels and sequence tags, matching deployment NLP outputs. However, the deployment additionally requires confidence scores, uncertainty flags, and ranked candidates — none of which are directly evaluated. The deployer's specified multi-label-with-confidence output behavior is not measurable with the current scoring framework [Q56, DEFT2021-D11].

OF-2: Not applicable — deployment is text-based with no speech output requirement.

OF-3: Not directly applicable — primary users (clinicians, nursing staff) have high domain expertise [elicitation Q2]; literacy is not a concern.

OF-4: External validity is harmed by absence of confidence calibration, uncertainty flagging, and ranked-candidate scoring (DEFT2021-D11, DIAMED-D10, DEFT2020-D16). Severe class imbalance in CAS, ESSAI, and PxCorpus (CAS-D21, ESSAI-D13, PXCORPUS-D15) means standard accuracy can mask poor minority-class performance on clinically critical correction intents and rare ICD-10 chapters. The benchmark's own observation that CamemBERT models produce only 'O' labels in low-resource scenarios [Q73] illustrates this mode of evaluation failure.

ID	Evidence
Q2	"We evaluate 8 state-of-the-art pre-trained masked language models (MLMs) on general and biomedical-specific data, as well as English specific MLMs to assess their cross-lingual capabilities." (p.1)
Q56	"The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions." (p.6)
Q73	"However, all the models based on CamemBERT face difficulties in corpora with limited amount of data, such as MantraGSC Patents, where they fail to generate labels other than 'O'." (p.7)
DIAMED-D10	Le traitement était constitué d'énoxaparine d'antiagrégant plaquettaire, de diurétique de l'anse, de dérivés morphiniques et d'oxygène — Multi-system case forced into single digestive label — masks multi-morbidity
DEFT2021-D11	[1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 1.0, 0.0, 1.0] — Strictly binary 0/1 label vector — no confidence or soft-label dimension present
ESSAI-D13	Chaque cycle dure 28 jours — Trivially neutral sentence — inflates neutral class count, contributing to severe label imbalance
PXCORPUS-D15	negate=11, replace=3 in 500 sampled — Severe class imbalance; majority-class baseline masks minority-class performance
DEFT2020-D16	Après inspection, elles ont toutes été exclues. — Annotator disagreement averaged away in scoring; calibration signal lost
CAS-D21	buffer: negation_speculation:6, negation:100, neutral:368, speculation:26 — Heavy class imbalance; neutral dominates at ~74%

Information Gaps

- Per-class F1 (not just macro/aggregate) and calibration curves are not reported in the benchmark for the deployment-critical minority classes

Requires Expert Verification

- What calibration thresholds (ECE, Brier score, AUROC per label) are operationally adequate for the deployer's clinician-adjudication workflow

Remediation

Gap 1: Standard F1/Spearman/SeqEval scoring with majority-class baselines cannot measure calibration, uncertainty flagging, or ranked-candidate behavior; severe class imbalance in CAS/ESSAI/PxCorpus masks minority-class performance

Recommendation: Augment the benchmark's reporting with per-class F1, calibration plots, and ECE for all classification tasks. For PxCorpus minority classes (negate, replace) and CAS/ESSAI non-neutral labels, report F1 explicitly stratified by class to surface clinically critical failure modes in prescription correction and negation detection.